

Smallest String With Swaps

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| Logic & Approach Notes

ye basically dsu ka question he

har ek pair basically ek group bana rha he

to initially sbko merge krke groups form krenge

phir leader->indices ka ek map

phir har leader ke indices pe jo characters honge wo store krenge

sort indices , sort characters

| C++ Implementation

```

class Solution {
public:
    class dsu {
    public:
        vector<int> parent;
        vector<int> size;

        dsu(int n) {
            parent.resize(n);
            size.resize(n, 1);
            for(int i = 0; i < n; i++) {
                parent[i] = i;
            }
        }

        int findparent(int node) {
            if(parent[node] == node) {
                return node;
            }
            parent[node] = findparent(parent[node]);
            return parent[node];
        }

        void unionit(int u, int v) {
            int pu = findparent(u);
            int pv = findparent(v);

            if(pu == pv) return;

            if(size[pu] < size[pv]) swap(pu, pv);

            parent[pv] = pu;
            size[pu] += size[pv];
        }
    };

    string smallestStringWithSwaps(string s, vector<vector<int>>& pairs) {
        // ye basically dsu ka question he
        // har ek pair basically ek group bana rha he
        // to initially sbko merge krke groups form krenge
        // phir leader->indices ka ek map
        // phir har leader ke indices pe jo characters honge wo store
krenge
        // sort indices , sort characters

        int n = s.size();
        dsu ds(n);
        for(auto &it : pairs) {
            int u = it[0];
            int v = it[1];
            ds.unionit(u, v);
        }
    }
};

```

```

unordered_map<int, vector<int>> mp;
// will store leader->followers

for(int i = 0; i < n; i++) {
    int boss = ds.findparent(i);
    mp[boss].push_back(i);
}

string ans = s;
for(auto &it : mp) {
    int boss = it.first;
    vector<int> indices = it.second;
    vector<char> ch;

    for(auto &idx : indices) {
        ch.push_back(s[idx]);
    }

    sort(ch.begin(), ch.end());
    sort(indices.begin(), indices.end());

    for(int i = 0; i < indices.size(); i++) {
        ans[indices[i]] = ch[i];
    }
}

return ans;
}
};

```